CSE 341: Microprocessors
Department of Computer Science and Engineering
Brac University

SET A

Examination: Final
Duration: 1 Hour 40 Minutes

Semester: Spring 24
Full Marks: 35

Name:	ID:	Section:

ANSWER ALL THE QUESTIONS

Question 1: [CO3]

a. A student suggested that the FFFFFh address can be accessed using variable I/O Address instructions. **Do you agree with this statement**? Justify your answer. [1]
b. To adopt the utility of **8 channels, estimate** the minimum number of secondary 8237 DMA controllers needed. **Show working [1]**
c. **Explain** the fundamental differences between isolated I/O and memory-mapped I/O in the context of the 8086 microprocessor architecture. **[2]**
d. **Compare** the three I/O Data transfer techniques in detail. **[3]**

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Question 2:[CO3]

a. **Explain** why 8259 PIC keeps an IRR register instead of forwarding the interrupt request signal directly to the priority resolver. **[3]**
b. After an ISR execution ends, the 8259 PIC must be informed that the interrupt has finished executing so that it can update the registers (IRR, ISR) accordingly. **What** problem will occur if the PIC is not informed about the execution of the ISR? **Explain** your reasoning. **[2]**
c. **Assume** that you have been given some faulty PICs that have IR3 and IR5 pins disabled. To adopt the utility of 25 peripheral devices, **determine** how many secondary PICs will be required. **[2]**

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Question 3: [CO3]

1. Suppose while the current instruction is executing, an NMI occurs. The NMI will be serviced before the next instruction begins executing. **Find** out the physical address of the NMI interrupt service routine. **[2]**
   Interrupt Vector Table(in hex): **BB 08 0B 02 65 04 70 00 16 05 DA 09 65 04 70 00**

2.

Address	00083h	00084h	00085h	00086h	00087h	185C0h	185C1h	185C2h
Data	9Ch	A0h	15h	02h	1Ah	50h	6Eh	78h

If the Intel 8086 encounters the INT 33 instruction, then **calculate** the starting address of the corresponding ISR. Also, **what** will be the first 2-byte hex instruction that will be fetched to the instruction queue from the ISR? **[2+1]**

3. Write two differences between software interrupts and hardware interrupts of 8086. **[2]**

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Question 4: [CO4]

a. Briefly **assess** how the real address mode differs from the protected virtual address mode of the Intel 80286. **[2]**

b. "The Execution Unit of the 8086 and the 80286 provides the same functionalities" - **Argue** with logic on whether this statement is true or false. **[2]**

c. "With the introduction of IOPL, not all tasks will have the privilege to perform input/output". Briefly **validate** this statement with an example. **[3]**

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Question 5: [CO2]

Suppose you have an Intel 8086 working at a frequency of 50MHz. Then, it comes across a memory-read instruction from an odd addressed location whose **Instruction cycle** would be 160ns.

a. Based on the given scenario, deduce the possible values of the A_0 and BHE' pins for the above given instruction. Please show the necessary calculations that support your conclusion. **[3]**

b. **Draw** the timing diagram for the given instruction. You must showcase the values of the A_0 - A_{19}, D_0 - D_{15}, ALE, M/IO, DT/R', and DEN' pins. **[4]**